

Everett Lopez

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EXPERIENCE

KAHLERT SCHOOL OF COMPUTING | UNDERGRADUATE

RESEARCH ASSISTANT

Aug. 2025 – Present | Salt Lake City, UT

- Engineered Python-based cryptographic workflows to support C2PA privacy-preserving media provenance research, implementing hashing algorithms and polynomial-commitment schemes.

PROJECTS

CHAT EXPRESS MESSAGING PLATFORM

GitHub, YouTube

FastAPI · SQLAlchemy · REST API Design · React · TypeScript

- Engineered the full backend architecture using FastAPI, including modular routers for accounts, chats, and messages, with clearly defined REST endpoints consumed by a React frontend.
- Designed and implemented SQLAlchemy data models (Account, Chat, Message) with relational mappings, validation, and efficient query patterns for real-time message retrieval.

SPOTIFY GENRE INTELLIGENCE DASHBOARD | Website

JavaScript · D3.js · Spotify Web API · Data Engineering

- Engineered a custom data ingestion and transformation pipeline to replace deprecated Spotify endpoints, generating consistent datasets across 10+ genres.
- Built backend logic for genre matching, subgenre expansion, and long-term trend analysis.

EXPLORATORY ANALYSIS OF FRAUDULENT BEHAVIOUR IN MOBILE TRANSACTIONS | Website

Python · Pandas · NumPy · Seaborn · Matplotlib · StatsModels

- Analyzed 6.3M+ simulated mobile money transactions to identify metadata-driven fraud patterns and behavioral anomalies.
- Built a full Python workflow to clean, structure, and model raw transaction logs using Pandas and NumPy, producing analysis-ready datasets.
- Applied Seaborn, Matplotlib, and StatsModels to visualize temporal trends, evaluate feature significance, and establish baseline fraud-detection signals.

EXTRACURRICULAR / CERTIFICATES

TATA GROUP DATA ANALYTICS JOB SIMULATION Virtual

Experience Program — Certificate of Completion

- Completed a structured data analytics simulation through Forage, modeling real-world tasks at Tata Group.
- Designed an AI-driven collections strategy leveraging agentic AI and automation, incorporating ethical AI principles, regulatory compliance, and scalable implementation frameworks.

EDUCATION

UNIVERSITY OF UTAH

B.S. SOFTWARE DEVELOPMENT W/

CERTIFICATE IN DATA SCIENCE

August 2021 - May 2027

College of Engineering

TECHNICAL SKILLS

BACKEND ENGINEERING

Python · FastAPI · REST APIs · SQL · MySQL · Data Modeling · API Architecture · AWS (EC2, cron) · Git

DATA ENGINEERING

Pandas · Numpy · Matplotlib · ETL Pipelines

FRONTEND

React · TailwindCSS · JavaScript · D3.js · HTML/CSS

LANGUAGES

Java · C# · C/C++

COURSEWORK

UNDERGRADUATE

Database Systems

Calculus I-III

Data Wrangling

Web Development

Software Practice I-II

Data Structures & Algorithms

REFERENCES

GARRETT GREINER

Research Mentor – Kahlert SOC

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LINKS

GitHub: [EverettLopez](#)

LinkedIn: [EverettLopezJr](#)

Website: [Everett's GitHub Page](#)